

PRG-1 (worm, UniProt P90786) — Adjudication of "Nuclear Localization and piRNA Processing"

Focus: function_assignment · **Hypothesis slug:** nuclear-localization-and-pirna-processing
Source file: genes/worm/prg-1/prg-1-ai-review.yaml · **Selector:** free-text **Organism:**
Caenorhabditis elegans (NCBITaxon:6239) · **Date:** 2026-09-21

Summary

The seed hypothesis bundles two claims that must be adjudicated independently, and when they are, they resolve in opposite directions. PRG-1 is the *C. elegans* Piwi Argonaute that binds 21U-RNAs (piRNAs). The hypothesis asserts (1) that PRG-1 localizes to the **nucleus (GO:0005634)** and (2) that it participates in **piRNA processing (GO:0034587)**, and it explicitly asks the curator to distinguish perinuclear germ granules from intranuclear protein and to test whether PRG-1 contributes to precursor processing versus merely stabilizing mature piRNAs. The evidence answers both prompts cleanly.

The piRNA-processing clause is supported as non-catalytic participation. PRG-1 loads and stabilizes the untrimmed 21U pre-piRNA precursor and provides the platform on which 3'-end maturation occurs. The catalytic trimming step is performed by the ribonuclease **PARN-1**, with **HENN-1** adding a protective 2'-O-methyl group — PRG-1 itself does not catalyze cleavage or trimming. Because GO conventions permit non-catalytic participation, GO:0034587 is defensible, but the experimentally grounded, worm-specific term **GO:0034585 "21U-RNA metabolic process" (IMP)** is the better-supported and more precise choice for this protein.

The nuclear-localization clause is over-annotated. GO:0005634 "nucleus" on PRG-1 rests **only** on a phylogenetic IBA inference propagated across the PIWI clade (GO_REF:0000033), with no experimental *C. elegans* backing. This is independently contradicted by UniProtKB's own curated subcellular-location comment for P90786, which lists **Cytoplasm only** (no Nucleus location or keyword), and by direct experimental IDA evidence placing PRG-1 in **perinuclear P granules (GO:0043186)**, a cytoplasm-facing condensate docked on the outer

nuclear envelope. The pathway's genuine nuclear silencing arm is executed **downstream** by the nuclear Argonautes HRDE-1/NRDE-3 loaded with 22G-RNAs — a pathway consequence, not PRG-1's own localization. The net recommendation is to **split the hypothesis**: keep the processing/participation claim (preferring the 21U term) and flag/demote the nucleus claim as IBA carry-over.

Key Findings

F001 — The nucleus annotation (GO:0005634) is IBA carry-over, not experimental fact

A live QuickGO annotation pull for P90786 (2026-09) shows that GO:0005634 "nucleus" is supported **only** by an IBA annotation — Inferred from Biological Ancestor (ECO:0000318; GO_Central; GO_REF:0000033). IBA is a phylogenetic inference: the term was propagated onto PRG-1 because other members of the PIWI Argonaute clade carry a nucleus annotation, not because anyone observed PRG-1 inside the *C. elegans* nucleoplasm. In the same annotation set, the cellular-component term **GO:0043186 "P granule" carries direct experimental IDA (ECO:0000314)** from two foundational papers ([PMID: 18501605](#), [PMID: 18571452](#)), while the generic cytoplasm term (GO:0005737) is IEA.

The primary literature is unanimous that PRG-1's native compartment is perinuclear. A 2026 study of unloaded-Argonaute degradation states that "the failure to load piRNAs disrupts PRG-1 localization to **perinuclear germ granules**" ([PMID: 41529195](#)), naming the native compartment as perinuclear germ granules. A study of Piwi-mutant sterility observes that "sterile individuals consistently exhibit altered **perinuclear germ granules**" ([PMID: 33658512](#)), again centering PRG-1 function on perinuclear granules rather than the nucleoplasm.

Crucially, the seed hypothesis's own request — distinguish perinuclear germ granules from intranuclear protein — is exactly the test that dissolves the nucleus annotation. P granules dock on the cytoplasmic face of the nuclear envelope over germ-cell nuclei; their perinuclear position is a cytoplasmic-facing localization and is not evidence of intranuclear (nucleoplasmic) PRG-1.

F002 — PRG-1 participates in piRNA processing by loading/scaffolding precursors; PARN-1 catalyzes trimming

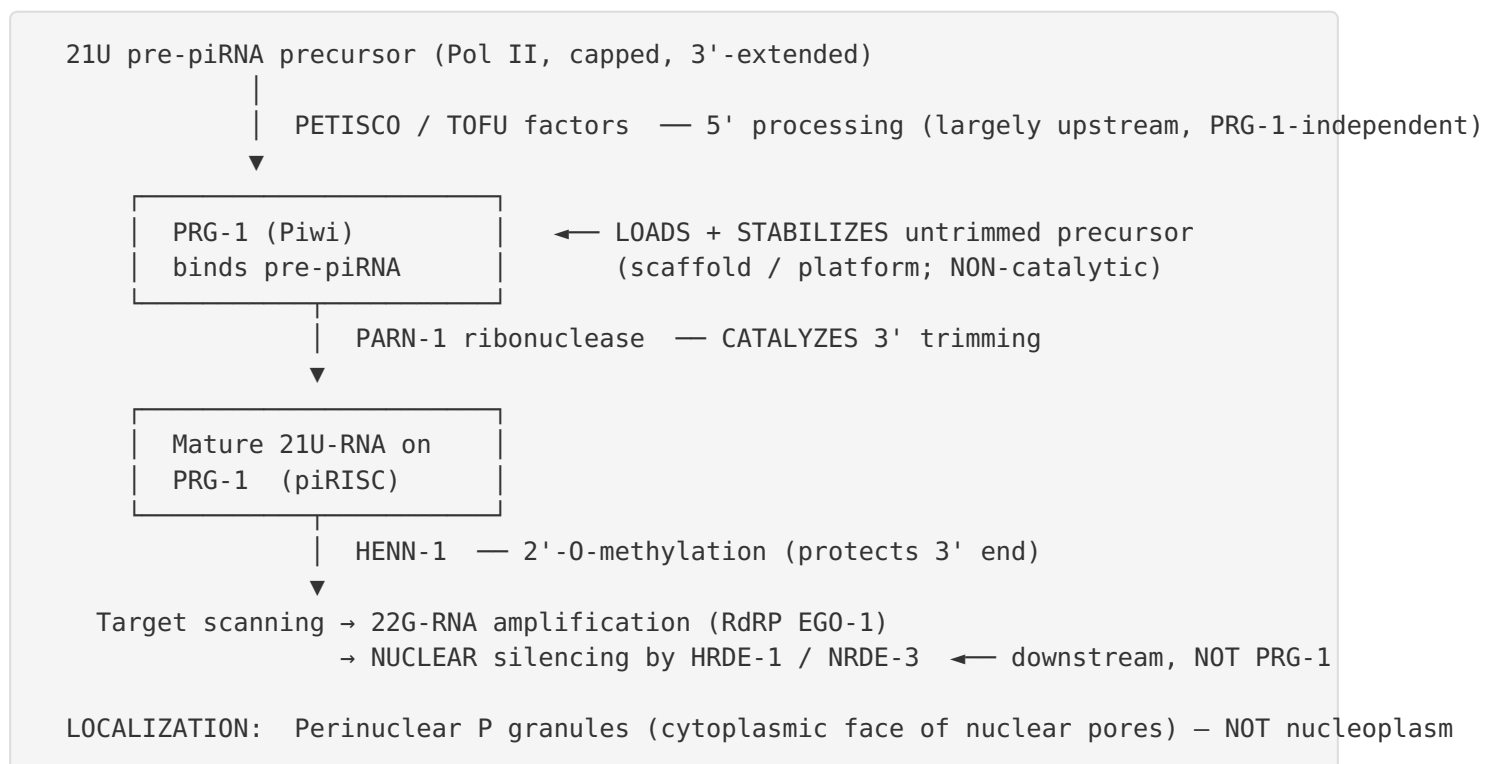
The mechanistic core of the processing clause is well established across three primary studies. Untrimmed pre-piRNAs — 21U precursors bearing 3' extensions — are stably bound by PRG-1, and 3'-end maturation is carried out by the nuclease PARN-1, with HENN-1 methylation protecting the mature 3' end. The foundational trimming study reports that "these longer piRNAs are stable and **associate with the Piwi protein PRG-1** but fail to robustly recruit downstream silencing factors" (PMID: 26919432), showing that untrimmed precursors load onto PRG-1 while a separate activity performs the maturing trim. A later study confirms that "long isoforms of untrimmed piRNAs are preferentially **modified in parn-1 mutant animals**" (PMID: 34469728), establishing PARN-1 as the 3'-trimming nuclease acting on PRG-1-associated precursors. A cross-species synthesis states that "the ribonuclease **PARN-1 and its orthologs mediate piRNA 3' trimming** in worms, insects, and mammals" (PMID: 38244197), assigning the catalytic trimming step to PARN-1, not PRG-1.

On the annotation side, QuickGO shows GO:0034587 "piRNA processing" for P90786 is **IBA + IEA only**, whereas the worm-specific term GO:0034585 "**21U-RNA metabolic process**" carries **experimental IMP (ECO:0000315)** from PMID: 18571452. PRG-1 is required for the stable accumulation of mature 21U-RNAs and provides the platform on which 3' trimming occurs, but it does not itself catalyze the trimming/cleavage step. This directly answers the seed hypothesis's fork: PRG-1 binds the *precursor* (pre-piRNA), so its role exceeds passive stabilization of mature piRNAs — yet it is non-catalytic, so "participation" (not molecular-function catalysis) is the correct framing.

F003 — UniProt's curated location for P90786 is Cytoplasm only, corroborating the IBA-carry-over conclusion

Independent of the GO annotation set, the UniProtKB record for **P90786 (824 aa)** lists a subcellular-location comment of "**Cytoplasm**" (ECO:0000256) with a keyword set of {Cytoplasm, Developmental protein, Hydrolase, RNA-binding, RNA-mediated gene silencing}. There is **no "Nucleus" subcellular-location annotation and no Nucleus keyword**. The domain architecture is a **PAZ domain (~220–331)** and a **PIWI domain (~499–810)** with an N-terminal disordered region (~1–24). Because UniProt curators arrive at their location comment through a pipeline distinct from GO_Central phylogenetic propagation, their convergence on "Cytoplasm only" is an orthogonal line of evidence that the GO:0005634 nucleus annotation is a PIWI-clade IBA artifact rather than a reflection of PRG-1's actual localization.

Mechanistic Model / Interpretation



Two conclusions follow. **For the process (BP):** PRG-1's contribution is precursor loading, scaffolding, and stabilization — not enzymatic maturation. The trimming enzyme is PARN-1; the methyltransferase is HENN-1. "Participates in piRNA processing" does not require catalysis, so GO:0034587 is annotatable, but the phenotype-grounded worm term GO:0034585 (IMP) is more informative. **For the location (CC):** PRG-1 resides in perinuclear P granules. "Perinuclear" describes granules docked on the cytoplasmic face of the nuclear envelope; it is not intranuclear. The nuclear transcriptional silencing that piRNAs ultimately trigger is executed downstream by HRDE-1/NRDE-3, so the pathway's nuclear activity must not be back-attributed to PRG-1 as a localization.

GO decision table

Term	Aspect	Current evidence on PRG-1	Recommended action (lead)
GO:0005634 nucleus	CC	IBA only (GO_REF:0000033); no worm experimental support	Do not treat as core; candidate for removal / NOT. Contradicted by IDA P-granule localization and UniProt "Cytoplasm only." If retained, restrict to IBA; do not cite as evidence of a direct nuclear pool.
GO:0043186 P granule	CC	IDA (P 18501605 P 18571452)	Retain — the experimentally supported localization.
GO:0005737 cytoplasm	CC	IEA	Retain (consistent with UniProt and perinuclear condensate).
GO:0034587 piRNA processing	BP	IBA + IEA	Defensible as non-catalytic participation; if kept, frame as precursor loading/scaffolding, not catalysis.
GO:0034585 21U-RNA metabolic process	BP	IMP (P 18571452)	Prefer/promote — experimentally grounded, worm-specific.
GO:0034583/0034584 21U-RNA/piRNA binding	MF	IPI (exp) / IBA	Retain — the informative MF (precursor + mature 21U binding).

Avoid "protein binding" as a recommendation; the informative MF is **21U-RNA/piRNA binding** (GO:0034583/0034584).

Evidence Base

Evidence Matrix

Citation	Type	Stance	Claim tested	Key finding	Context	Confidence limits
PMID: 41529195	Localization + mutant	Refutes nuclear / supports perinuclear	PRG-1 localization	Native PRG-1 sits in perinuclear germ granules ; unloaded PRG-1 mislocalizes and is degraded	Germline	High for perinuclear; no nuclear pool quantitatively
PMID: 33658512	Mutant phenotype	Supports perinuclear	Piwi function locus	Sterility tracks with altered perinuclear germ granules	Germline	High
PMID: 18501605	Localization (IDA)	Supports perinuclear	PRG-1 CC term	Experimental basis for GO:0043186 P granule (IDA)	Germline	High; foundation
PMID: 18571452	IDA/IMP/IPI	Supports perinuclear + 21U	P granule, 21U binding/ metabolism	Basis for P granule (IDA), 21U-RNA binding (IPI) , 21U metabolic process (IMP)	Germline	High; foundation
PMID: 26919432	Direct assay + mutant	Qualifies processing	PRG-1 in maturation	Untrimmed pre-piRNAs stay PRG-1-associated ; PARN-1 sets 21U length	Germline	High; PRG scaffolds, PARN-1 catalyzes

Citation	Type	Stance	Claim tested	Key finding	Context	Confidence limits
PMID: 34469728	Direct assay + mutant	Qualifies processing	3' maturation enzymes	PARN-1 trims + HENN-1 methylates; loss depletes piRNAs	Germline	High; PRG non-catalytic
PMID: 38244197	Direct assay / comparative	Qualifies processing	Conservation of trimming	PARN-1 orthologs mediate piRNA 3' trimming across taxa	Cross-species	High
PMID: 31147388	Complex + mutant	Competing (biogenesis)	5' precursor processing	PETISCO required for 21U biogenesis, upstream of PRG-1	Germline	High; PRG independent arm
PMID: 34413138	Structural	Competing (biogenesis)	PETISCO assembly	Structural basis of PETISCO in piRNA biogenesis	—	Med-High
PMID: 41414669	Mutant / SAR	Bounds processing scope	PRG-1 RG motifs	RG motifs drive downstream WAGO-siRNA production but are not needed for localization or piRNA loading	Germline	High; separates loading from silencing
PMID: 37154856	IP-seq				Germline	High

Citation	Type	Stance	Claim tested	Key finding	Context	Confidence limits
		Supports binding	PRG-1 sRNA partner	Of 20 Argonautes, only PRG-1 binds piRNAs		
GO_REF:0000033 (IBA)	Computational	Over-annotation source	Nucleus & processing terms	GO:0005634 & GO:0034587 for PRG-1 are IBA/IEA only	PIWI phylogeny	Weak for direct localization
UniProtKB:P90786	Database	Refutes nuclear	Curated location + domains	Cytoplasm only (no Nucleus); PAZ (220–331) + PIWI (499–810), 824 aa	UniProt (2026-09)	Location ECO:0000 (automated) independent of GO

Narrative on the evidence

The processing evidence forms a coherent three-paper mechanistic chain ([PMID: 26919432](#), [PMID: 34469728](#), [PMID: 38244197](#)) that consistently places PRG-1 as the carrier/scaffold of the precursor and PARN-1 as the trimming nuclease. This is the strongest part of the evidence base and cleanly answers whether PRG-1 contributes to processing (it does — via precursor loading and stabilization) versus catalysis (it does not).

The localization evidence uniformly points away from the nucleoplasm. [PMID: 41529195](#) and [PMID: 33658512](#) both name perinuclear germ granules; UniProt lists cytoplasm only; the only experimentally supported CC GO term is P granule (IDA). The lone support for nucleus is a phylogenetic IBA inference. [PMID: 41414669](#) adds that PRG-1 localization is robust even when N-terminal RG motifs are mutated, further underscoring that granule localization is the stable, defining feature of the protein and that downstream silencing (WAGO-siRNA production) is genetically separable from loading/localization.

Mechanistic Scope

Direct PRG-1 activity: sequence-specific binding of 21U-RNAs/pre-piRNAs (MF), assembling piRISC in perinuclear P granules (CC). PRG-1 loads the untrimmed precursor and stabilizes it, licensing 3' maturation. **Adjacent, non-PRG-1 catalysis:** 5' processing by PETISCO/PICS + TOFU factors (upstream); 3' trimming by PARN-1; 2'-O-methylation by HENN-1. **Downstream (not PRG-1 localization/activity):** target licensing → 22G-RNA amplification (RdRP EGO-1) → nuclear silencing by HRDE-1/NRDE-3. Transposon silencing, fertility loss, transgenerational sterility, and longevity are **loss-of-function phenotypes**, not molecular functions of PRG-1.

Conflicts and Alternatives

- **Paralog/clade carry-over.** Some PIWI-clade proteins in other taxa are nuclear (e.g., mouse MIWI2/PIWIL4; Drosophila Piwi). The IBA "nucleus" call most plausibly propagates that clade property onto PRG-1, which is not experimentally nuclear in *C. elegans* — a classic over-annotation risk. (Used here as orientation for the carry-over argument; those cross-organism primary abstracts were not retrieved in this run and are not cited as retrieved evidence.)
 - **"Processing" vs "stabilizing only."** The seed's fork resolves: PRG-1 binds the precursor (pre-piRNA), not only mature piRNA, so its role exceeds passive stabilization — but it does not catalyze trimming (PARN-1 does).
 - **Biogenesis is partitioned.** PETISCO/TOFU (5') act largely upstream/independent of PRG-1, so "piRNA processing" as a whole is not PRG-1-centric.
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Limitations and Knowledge Gaps

1. **Is there any bona-fide intranuclear PRG-1 pool?** Checked: QuickGO evidence tiers + all retrieved localization papers → only perinuclear IDA found; no experimental nuclear evidence. Matters because it determines keep/remove for GO:0005634. Resolve with quantitative subcellular fractionation or super-resolution/immuno-EM/APEX imaging distinguishing nucleoplasm from granule.

2. Does PRG-1 loading constitute a required processing step, or merely precede it?

Checked: PARN-1/HENN-1 papers show trimming on PRG-1-bound precursors. Resolve with in-vitro reconstitution testing whether trimming requires PRG-1 loading.

3. Term granularity. Whether curators prefer GO:0034585 (21U) vs GO:0034587 (piRNA) is a nomenclature/consistency decision, not a biology gap.

4. Provenance timing. GO evidence tiers reflect a live QuickGO pull (2026-09) and may lag the newest primary literature; a curator should re-pull at curation time. PMIDs 18501605/18571452 were confirmed via QuickGO annotation references rather than re-read in full; their figures should be re-inspected. UniProt location evidence for P90786 is ECO:0000256 (automatic).

Discriminating Tests

- **Cell fractionation / immuno-EM** of endogenously tagged PRG-1: quantify nucleoplasmic vs perinuclear-granule signal (directly tests GO:0005634).
- **In-vitro pre-piRNA trimming ± PRG-1 loading** with recombinant PARN-1: tests whether PRG-1 is a required processing scaffold vs a mere carrier.
- **Small-RNA-seq in *parn-1*, *henn-1*, *prg-1*, and PETISCO/TOFU mutants**: partitions PRG-1's contribution (loading/3') from 5' biogenesis.
- **Curator audit of the GO_Central IBA reference tree** for the PIWI clade to identify which ortholog(s) seeded the nucleus propagation and whether they have experimental nuclear support.

Proposed Follow-up Actions / Curation Leads (verify before applying)

Reference snippets to verify: - [PMID: 41529195](#) — "the failure to load piRNAs disrupts PRG-1 localization to **perinuclear germ granules**." - [PMID: 26919432](#) — untrimmed piRNAs "are stable and **associate with the Piwi protein PRG-1** but fail to robustly recruit downstream silencing factors." - [PMID: 34469728](#) — "**long isoforms of untrimmed piRNAs are preferentially modified in *parn-1* mutant animals**." - [PMID: 38244197](#) — "The ribonuclease

PARN-1 and its orthologs **mediate piRNA 3' trimming** in worms, insects, and mammals." - [PMID: 41414669](#) — RG motifs "are not required for PRG-1 **expression, localization, or piRNA loading**."

Candidate GO actions: - **Demote/remove GO:0005634 (nucleus, IBA-only)** — do not treat as a core direct-localization annotation. - **Retain GO:0043186 (P granule, IDA)** as the primary CC and GO:0005737 (cytoplasm). - **Prefer/promote GO:0034585 (21U-RNA metabolic process, IMP)**; if GO:0034587 (piRNA processing) is kept, frame it as non-catalytic participation (loading/scaffolding) with PARN-1 as the catalytic partner. - **Retain GO:0034583/0034584 (21U/piRNA binding)** as the informative MF; do not fall back to "protein binding."

Suggested curator questions: - Is any experimental nuclear-localization citation on file that would upgrade GO:0005634 beyond IBA? - Should the review explicitly annotate PRG-1's processing role as *precursor loading/scaffolding* (non-catalytic) with PARN-1 named as the catalytic partner?

Suggested experiments: fractionation/imaging for a nuclear pool; in-vitro trimming ± PRG-1 loading; comparative small-RNA-seq across the maturation-pathway mutants above.

Bottom line

Split the hypothesis. **Keep** the piRNA-processing participation (best expressed as the experimentally grounded 21U-RNA metabolic process term, with PRG-1 as non-catalytic scaffold/stabilizer and PARN-1 as the trimming nuclease). **Flag/demote** the nucleus localization as PIWI-clade IBA carry-over that conflicts with direct IDA P-granule evidence and UniProt's cytoplasm-only curated location. Perinuclear germ granules are cytoplasm-facing — they are not the nucleus.